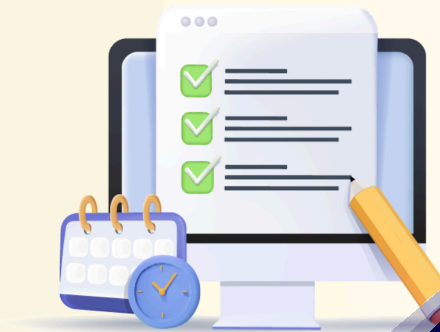


POSTGRADUATE - MATH1318 Time Series Analysis - Final Project (Written report and presentation instructions and written report submission)

Assessment 3

Final Project



Course Code: MATH1318

Weighting: This assessment is worth 50% (45% Written Report + 5% Presentation) of your final mark for this Course

Length: No word or page limit

Task approach: Individual or Team

Presentation due date and time: Friday the 30th of May, Week 12, 03:00 pm AEDT

Written report due date and time: Sunday the 15th of June, 11:59 pm AEDT

Feedback: Feedback will be provided via Email and real-time in class

Please click or tap on the relevant sections to find out more:

▼ 1. Course Learning Outcomes Assessed

This assessment supports the following learning outcomes:

- CLO1. Present time series in an informative way, both graphically and with summary statistics;
 - CLO2: Develop stationary and non-stationary, and seasonal and non-seasonal time series models;
 - CLO3. Estimate model parameters and compare different models developed for the same dataset in terms of their estimation and prediction accuracy;
 - CLO4. Prepare both oral and written reports to present the results of time-series analyses.
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▼ 2. Overview of Assessment

The purpose of this assignment is to prepare a presentation and an analysis report using real data similar to workplace tasks. You will showcase your time series modelling and forecasting skills along with R software and reporting skills in this assessment task. This assessment resembles the tasks you would be doing in industry. Therefore, put your best effort into the assessment and use the feedback given by the course coordinator to improve your performance for the next assessments.

▼ 3. Assessment Task

Instructions: To-Dos

Please read the instructions first and then watch [the video here](#) for more explanation.

You can work on this assignment in teams of **3** students. But if teaming up with your peers is impossible, you can work on the project individually, too. In this case, please send the course coordinator an email to let them know. Teamwork in this assessment task aims to get students to practice teamwork skills, one of the important real-life skills. Therefore, it is recommended that teams be built up for the project. However, since teamwork is optional in this assessment, you can also work individually. Please see the RMIT resource for teamwork [here](#). .

Undergraduate students can team up with only undergraduate students; the same rule applies to postgraduate students.

In this assignment,

- you will find a time series dataset, preferably a public one,
- set an achievable research question (for example, if you are working with wind speed data, your research question can be "What are the most accurate forecasts for the wind speed at a location for the next 10 days?"),

- analyse the data by using the analysis methods covered in the MATH1318 Time Series Analysis course this semester,
- accurately predict the series for the next 10 units of time (if your series is annual, 10 units of time = 10 years; if your series is monthly, 10 units of time = 10 months, and so on) to give a conclusion on your research question and
- prepare a comprehensive analysis report including descriptive analysis, proper visualisation, model specification, model fitting and selection, and diagnostic checking and interpretations of the results.

The data quality will be assessed in terms of the study's originality and the difficulty of modelling. Therefore, the length and characteristics of the time series you choose for the final project are just up to you.

All final project reports submitted will be presented by only one of the team members during the final project presentations.

Each report needs to have a cover sheet. The template of the cover sheet is given [here](#). All team members need to contribute equally to the project and ensuring this is their responsibility. You need to declare each team member's contribution to the project via the cover sheet. If all team members contribute equally, which is the expected situation, put $1/(\text{The number of team members})$ for each team member under the **"Contribution to the project"** column. For example, If you have a team of 3 and all team members have contributed equally, put $1/3$ for each team member. **All team members should watch [this video](#), which shows an example of bad practice for teamwork. By establishing a team and committing to do a team project instead of an individual one, all team members are assumed not to make the mistakes in this video. If you see any possibility that you will be in such a situation or will not contribute to the project equally for any reason, it is best to do the project individually.**

You will prepare a report that includes proper sections such as Introduction, Conclusion, Appendix and a chapter for each task. The report's structure is up to you as long as it is well-structured and neat. The report expected here is **not** a short business report or **not** an essay. You can refer to the RMIT resource [here](#)  for further information on report writing. Specifically, "Science Report" under "Sample report structures" in the link [here](#)  may be a good example of the report expected.

As in other data science competitions, e.g. datathon, this task will be like a real data science consulting task. So, I will not answer specific questions about data analysis or reporting. Instead, it is expected that you ask yourself what are the elements of a suitable and successful data analysis and how you might go about presenting your results in a written report format and as an oral presentation. I'm happy to talk about general questions on this task.

In this assessment task, you can use AI tools to assist you. All use of AI tools and their outputs should be appropriately acknowledged and referenced.

If you utilise a generative AI tool for any part of your analysis or report, you must have a separate section to explain

- which tool you used,
- how you used it for your analysis or report, and
- how did you ensure that the results of the AI tool are correct.

▼ 4. How to register/make changes to your team?

Please use the form [here](#) ➡ to register your team or make changes to your already registered team. **Only one of the team members** needs to register the team on behalf of all members. The students working individually also need to register as a team of one.

▼ 5. Presentation Instructions

You can either go for a **Video presentation** or a **Class presentation, either face-to-face in class or online over Collaborate Ultra**.

You can find information about oral presentations [here](#). ➡

Here are the rules for the **class presentations**:

- Each presentation will be limited to **3 to 5** (to be confirmed based on the number of groups going for a class presentation) minutes maximum. This would be increased based on the final number of class presentations.
- You can use **7** slides maximum, including the cover slide.
- For example, you can arrange your presentation such that
 - 2 slides for data description and visual analysis,
 - 1 slide for model specification,
 - 1 slide for parameter estimation,
 - 1 slide for residual analysis, and
 - 1 slide for the conclusion.
- Animations or sounds on slides are not allowed.
- The class presentations will take place over Collaborate Ultra on the date/time as announced in the Syllabus.
- The order of class presentations will be randomised.

This will be a good experience for explaining your analysis results within a limited time frame since you probably won't have a manager/supervisor who can listen to you for more than a couple of minutes in the outside world.

As for the **video presentations**, we have the following rules:

- Each presentation is limited to **5** minutes maximum.
- You can use a maximum of 7 slides, including the cover slide.
- **You do not need to do a class presentation if you submit a video presentation. You cannot submit a video presentation if you do a class presentation.**
- Your presentation must be in a common video format that I can open it.
- **You need to upload your presentation to a platform like YouTube, obeying the academic integrity rules, and submit the link via Canvas (here). This link is also available by following Canvas --> Assignments --> Project Presentation Submission.**

Please follow the course coordinator's announcements for more details.

▼ 6. Timeline

To help you prepare for the assessment, please consider the following activities:

Timeline	Activities
Weeks 1-8	• Search for a suitable open dataset • Decide if you will do the final project individually or in a team • If you are doing the project in a team, search for suitable team members from your degree group • Review Modules 1 to 5 • Start descriptive analysis of your dataset • Explore ARIMA modelling on your dataset
Week 9	• Review Modules 1 to 8 • Complete descriptive analysis • Explore SARIMA modelling on your dataset • Start preparing the project presentation • Explore Module 9

Timeline	Activities
Week 10	• Review Modules 1 and 9 • Complete descriptive analysis and ARIMA/SARIMA modelling • Explore GARCH modelling on your dataset • Work toward completing project presentation
Week 11	• Finish project presentation • Start preparing the written report
Week 12	• Finish and present project presentation • Get the Course Coordinator's feedback
Week 13 - 14	• Incorporate the Course Coordinator's feedback in your analysis • Finish the written report

▼ 7. Assessment submission and feedback

- All reports must be submitted via **Turnitin using the link/button at the bottom of this page.**
- Your submission should be uploaded as a PDF or Microsoft Word file.
- Your report should meet English language requirements.
- **You need to ensure that your report's similarity is less than 30%, according to the Turnitin similarity report.**

▼ 8. Assessment Criteria

Marking Rubric:

The rubric for MATH1318 is available [here](#) ↓ .

▼ 9. Academic Integrity

You are free to discuss the main aspects of the assignment with your classmates. However, keep in mind that this is an individual assignment, and you should demonstrate your own effort and understanding. Because assignments will be submitted through Turnitin, all the material you submit will be checked for plagiarism. If plagiarism is detected, both the copier and the student copied from will be responsible. Therefore, it is your responsibility to ensure you do not copy or do not allow other classmates to copy your work. You should ensure you understand your responsibilities by reading the [RMIT University website on academic integrity](#).

Everyone's understanding of academic integrity would differ. Therefore, I highly recommend you do the [Academic Integrity](#) credential to be on the same page. You will not get any additional marks or points, but you will improve your understanding and skills towards academic integrity when this credential is completed.

Please note that the dataset given for this assignment is totally different from those given in previous semesters, even though the story behind it is similar. Therefore, you must work on the data from scratch. Any solution to previous years' assignments will not work for this assignment. In that sense, if you have done this course before and intend to submit your previous report, please be aware that it is counted as copying from yourself and violates the academic integrity policy, and it will not be a solution to this assignment.

[Study Support Hub](#): Academic skills advisors and librarians will help you with researching your topic, finding information, improving your academic writing skills, academic integrity, effective learning strategies, statistics, and more!

[Academic Integrity/plagiarism](#): You can achieve academic integrity by honestly submitting work that is your own. Presenting work that fails to acknowledge other people's work within yours can compromise academic integrity.

[Special consideration](#): If unexpected circumstances beyond your control have affected your ability to complete an assessment, you may be eligible to apply for special consideration.

▼ 10. Extensions and Special Consideration

All the courses in your program follow the RMIT University Assessment policy for extensions and special consideration. Ensure you understand [these guidelines](#) before applying.

If you are prevented from submitting an assessment on time by circumstances outside your control, you may apply for an extension of up to seven (7) calendar days in writing (email) to your Course Coordinator **one working day BEFORE the due date**.

The extension requests without a supporting evidence document will be rejected. So, it is crucial to include an evidence which can be a photo of your roster, a medical certificate, itinerary, etc.

Examples of the type of circumstances that may be considered include:

1. unexpected short-term physical or mental illness;
2. primary carer responsibility for a family member with an unexpected illness;
3. an unexpected, unavoidable employment commitment;
4. other unexpected personal circumstances outside your control, such as bereavement, being the victim of a crime or other trauma, severe disruption to living arrangements, or financial hardship, for instance, the sudden loss of employment or income.

Please note that since the students need to do a minimum number of courses per semester, having another assignment with a close due date does not provide valid grounds for an extension.

Falsifying documents is considered fraud and treated very seriously. If you are proven to have falsified supporting documents for ANY REASON, you are likely to have your course result converted to fail or have other penalties applied. As a student, you must be aware of your obligations and responsibilities under [RMIT Student Conduct](#).

Your Course Coordinator will notify you (via email) that your extension has been approved and indicate the new due date if applicable.

If you require an extension of time longer than 7 days, you must apply for Special Consideration. An application for special consideration is made in advance of an assessment wherever possible, but normally within five working days after the assessment date.

To request an extension, please submit it through the "Assignment Extensions" form from the course navigation panel on the left **at least one working day before the assignment's due date**.

▼ 11. Late Submission of Assessment

Late submissions without an approved extension or special consideration will incur a 10% penalty per day applied to the assessment mark. An assessment submitted more than FIVE days after the deadline will be given a mark of 0% for that assessment.

For more information, please consult the course guide or contact your course coordinator.

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